Noroff
School of technology
and digital media

BACK-END DEVELOPMENT 2

Vocational Education

Noroff Fagskole AS

Professional Degree

Prog. Code	Bed2	**NQF Level**	5.2	**Credits**	60
Valid for	H23-V24	**Dated**	15.06.23	**Version**	1.4
Programme Manager	Tom Drange	**Line Manager - Campus**	N/A	**Line Manager - Online**	FJ Botha
Learning Environment	Campus	Full-Time		**Semesters**	2
	Online	Full-Time / Part-Time		**Semesters**	2 / 4

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About the programme

Back-end developers play an important role in any organisation or business that wants to offer effective digital services and updated server-based solutions. The services can require development against front-end solutions or independent work as a back-end developer.

The knowledge, skills and general competence gained from this study program will enable students to complete different industry-related tasks. They will be able to work in national and international companies needing back-end or full-stack developers. The need for this competency is increasing. Candidates can apply for jobs in businesses that are involved in developing internal solutions or in specialised software companies that tailor solutions for their customers.

This program builds on the Back-end 1 program and provides a more advanced technical competence within programming. The students will gain supplementary skills that will make them ready to work independently and as productive software development team members. Students will go in depth in more complex topics such as advanced functions in JavaScript, databases, cloud-based services, .NET and Microsoft systems. In addition, the students will gain the skills necessary in current front-end technologies to work as full-stack developers.

Learning Environment

The digital classroom
All students at Noroff have access to a digital classroom, referred to as the learning platform. Here the student can access relevant academic and practical information about the study programme. The learning platform also contains learning content, activities, delivery deadlines, work requirements and assessments for every course.

Online
Online studies are flexible since students can study from anywhere and at their own pace according to the academic progression and scheduled deadlines. Students access their learning material for each course through the learning platform, and discussion forums and/or chat groups are used for communication between fellow students and teachers.

Campus
As part of the campus community, students will have access to on-site teachers, guest lecturers, and other students during their learning journey. Campus students study in modern working environments and have access to professional equipment for practical training.

After graduation

Vocational education at Noroff can expand career opportunities and lay lifelong learning foundations. Throughout the programme, students will familiarise themselves with key competencies relevant to industry employment.

Career opportunities
After graduation, the candidate is qualified for employment as:

- Back-end JavaScript developer
- .Net developer
- Full-stack developer
- Generalist JavaScript Web developer

Programme Learning Outcomes

The Norwegian Qualifications Framework for lifelong learning (NQF) defines the levels of qualifications in the Norwegian educational system. These levels describe what a learner knows, understands, and can do as a result of a learning process. Categories in NQF are defined as:

Knowledge: Understanding theories, facts, principles, procedures in the discipline, subject area and/or occupation.
Skills: Ability to utilise knowledge to solve problems or tasks (cognitive, practical, creative and communication skills).
General Competence: Ability to independently utilise knowledge and skills in different situations.

After graduation from this programme, students have acquired the following learning outcomes:

Knowledge
The candidate:

- has knowledge of industry-relevant concepts, theories, models, processes and tools that are used in Back-end software development
- has knowledge of modern features of JavaScript programming language that are used in Back-end development
- has knowledge of services, processes and tools relevant to database technologies and cloud services
- has knowledge of theories and industry-relevant approaches that are used in data warehousing to analyse and report data
- has knowledge of industry-relevant Front-end and Back-end frameworks that are used in Full-stack application development
- can assess own work with databases and cloud services in relation to information security and data privacy regulations
- is familiar with the history, traditions, distinctive nature and place in the society of Back-end software development
- has insight into own opportunities for development in planning, designing and creating Back-end software solutions
- has insight into own opportunities to become a Full-stack software developer using JavaScript frameworks, libraries and tools

Skills
The candidate:

- can explain vocational choices of tools, methods and techniques used in planning, developing and design of Back-end solutions
- can explain vocational choices of database technologies and cloud-based services relevant to software and Back-end development
- can explain vocational choices of Front-end and Back-end frameworks used in Full-stack application development
- can explain vocational choices in data warehousing system building and data analysis and reporting
- can reflect over the practice as a Back-end developer and adjust it under supervision
- can reflect over choices in integrating cloud-based technologies
- can find and refer to information and vocational material and assess its relevance to Back-end and software development
- can find and refer to information and vocational material and assess its relevance to integrating cloud services into existing solutions
- can find and refer to information relating to information security regulations, best practices and their relevance to Back-end solutions

General Competence

The candidate:

- can plan and carry out vocational tasks and projects alone or as part of a group and in accordance with ethical requirements and principles in the field of Back-end development
- can plan and carry out software solutions in JavaScript alone or as part of a group and in accordance with ethical requirements and principles for software development
- can exchange points of view with others with a background in programming, Back-end and Front-end disciplines and participate in discussions about the development of good developer practice
- can contribute to organisational quality assurance, streamlining and optimisation both internal and external through Back-end development practices

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Course Overview

Cloud Services				FI2BECS75			
Credits	7.5	Weeks	4	Hours	168	Semester	1

Database Technologies				FI2BEDT75			
Credits	7.5	Weeks	4	Hours	168	Semester	1

Data Warehousing				FI2BEDW05			
Credits	5	Weeks	4	Hours	168	Semester	1

Semester Project		FI2BEP175					
Credits	7.5	Weeks	4	Hours	168	Semester	1

Information Security				FI2BEIS25			
Credits	2.5	Weeks	2	Hours	84	Semester	2

JavaScript in Practice				FI2BEJP75			
Credits	7.5	Weeks	5	Hours	210	Semester	2

Back-end Technologies				FI2BEBT75			
Credits	7.5	Weeks	6	Hours	252	Semester	2

Front-end Technologies		FI2BEFT75					
Credits	7.5	Weeks	5	Hours	210	Semester	2

Exam Project		FI2BEP275					
Credits	7.5	Weeks	5	Hours	210	Semester	2

Teaching and Learning

Noroff offers an engaging and student-active learning experience that prepares candidates for professional working life through unique and industry-relevant teaching and learning activities governed by the current learning outcomes. Teaching and learning engage students in learning by promoting a holistic understanding of the different issues and challenges relevant to the subject areas. Students will develop lifelong learning skills by fostering critical thinking, creativity, collaboration, and communication.

Activities can vary for campus and online delivery and are composed of theoretical and practical approaches, providing students with the best possible outcome for each course. Noroff distinguishes between teacher and student-led activities. Both are equally important and tailored to each course's educational approach. Teaching and Learning activities used in the courses are outlined in the course descriptions. More information about Noroff Teaching and Learning is available here: https://noroffquality.com/en/teachinglearningnoroff_en/

English is the primary language for teaching and assessments in this program.

Work Requirements and Assessment

Assessment impacts the student's learning significantly and concludes if the student has achieved the intended learning outcome and, if so, at what level. Assessments include summative and formative methods depending on the content of the learning outcome of each course.

A course usually consists of one or more work requirements. The most common is compulsory course assignments that assess the acquired competencies outlined in the course learning outcomes. Course assignments are assessed as Passed/Failed or graded from A to F, after which verbal or written feedback is provided. Tests can also evaluate students' achievements and are usually used in combination with compulsory assignments.

Online studies may also require students to deliver one or more compulsory module assignments during a course. This is to follow up and support the online students' learning path. Module assignments can be used as learning activities for campus students.
Work requirements and assessment methods for each course are described in the course descriptions. More information about assessment methods is available here: https://noroffquality.com/en/teachinglearningnoroff_en/

Documentation
The results are announced on the learning platform when coursework requirements and final assessments are completed and assessed. After completing and passing the programme, the student is issued a diploma.

Re-Examination
It is possible to re-sit all courses according to the guidelines clearly outlined in the school's regulations and procedures. This is defined in the quality handbook.

Equipment Requirements

Information about equipment requirements is available here: https://www.noroff.no/studier/fagskole/backend-utvikling

Online students are required to purchase and maintain their equipment.

Admission Requirements

There are two ways to meet the admission criteria and be enrolled as a student:

1. Professional Degree Diploma – Back-end Development 1 – Level 5.1 in NQF

2. Prior learning and work experience
   a. Prior learning and work experience mean knowledge, skills and general

competence that a person has acquired through

i. Formal learning, such as foreign education, higher education, folk high school or adult education
ii. Non-formal learning, such as courses or other structured training that is not part of the education system
iii. Informal learning, such as care work, organisational experience, leisure activities, volunteer work or work experience.

Applicants wishing to be recognised for prior learning or work experience must be at least 23 years of age.

More Information about admission requirements is available here: https://www.noroff.no/sok/opptakskrav

Course Descriptions

Course 1: Cloud Services

Code	FI2BECS75	**NQF Level**	5.2
Credits	7.5	**Weeks**	4
Course Leader		N/A	
Hours Per Week		42 Hours	
Teacher-led hours		64	
Student-led hours		104	
Total Hours		168	

About the Course

This course teaches candidates how to integrate cloud services into Back-end solutions effectively. Different cloud providers are discussed, focusing on their most widely used offerings as they relate to Back-end development. Candidates are provided with the knowledge of the tools and processes needed to utilise these services. The course also provides insights into using container-based technologies such as Docker and Kubernetes.

Cloud services have become ubiquitous in the field of software development, and knowing the various providers and services and the skills to integrate them into existing software projects has become essential. This course provides the necessary overview to successfully utilise cloud services in a software project.

Course Learning Outcomes

Knowledge

The candidate:

- has knowledge of industry-relevant cloud service providers
- has knowledge of the tools and processes used to integrate cloud services into software solutions
- has knowledge of the tools and techniques used to create and manage containerised software solutions

Skills

The candidate:

- can explain choices of tools used to integrate cloud services into software solutions
- can explain choices of tools and techniques used to create and manage containerised software solutions
- can find and refer to information and vocational material and assess its relevance with regard to cloud providers

General Competence

The candidate:

- can exchange points of view with others with a background in the trade/discipline and participate in discussions about the role of cloud providers in delivering cloud services
- can plan and carry out tasks related to integrating and managing cloud services in software solutions
- can plan and carry out tasks relevant to implementing and managing containerised software solutions

Teaching and Learning - Campus

In this course, the following teaching and learning methods may be used on campus:

Lecture	Online Lessons
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online

In this course, the following teaching and learning methods may be used online:

Online Lessons	Lecture
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments

This is a list of requirements to pass the course:

Course Assignment	Pass/Fail

Course 2: Database Technologies

Code	FI2BEDT75	**NQF Level**	5.2
Credits	7.5	**Weeks**	4
Course Leader		N/A	
Hours Per Week		42 Hours	

Teacher-led hours	64
Student-led hours	104
Total Hours	168

About the Course

This course aims to teach candidates to utilise various database technologies related to Back-end development. Candidates are taught how to write real-world, performant SQL queries. Non-relational databases are introduced and contrasted with relational databases. The candidate is also given the essential knowledge of what is needed to implement and interface with a cloud-based database.

Knowledge of a broader spectrum of database technologies, tools and paradigms will strengthen the candidates understanding of the concept of databases and provide a good foundation to effectively integrate them into future projects. Candidates will also have the capacity to critically analyse whether certain technologies are applicable for use in a particular software project.

Course Learning Outcomes

Knowledge

The candidate:

- has knowledge of techniques and tools used to write complex SQL queries
- has knowledge of the processes and tools utilised to interact with non-relational databases
- has knowledge of the services and APIs used to deliver cloud-based database services

Skills

The candidate:

- can explain choices of use of techniques and tools used to write complex SQL queries
- can explain choices of processes and tools used to interact with non-relational databases
- can explain choices of services and APIs used to integrate cloud-based databases

General Competence

The candidate:

- can plan and carry out tasks relating to implementing complex database queries and utilising non-relational databases
- can plan and carry out tasks linked to configuring and utilising cloud-based databases

Teaching and Learning - Campus

In this course, the following teaching and learning methods may be used on campus:

Lecture	Online Lessons
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online

In this course, the following teaching and learning methods may be used online:

Online Lessons	Lecture
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments

This is a list of requirements to pass the course:

Course Assignment	Pass/Fail

Course 3: Data Warehousing

Code	FI2BEDW05	**NQF Level**	5.2
Credits	5	**Weeks**	4
Course Leader		N/A	
Hours Per Week		42 Hours	
Teacher-led hours		64	
Student-led hours		104	
Total Hours		168	

About the Course

This course teaches candidates the fundamental knowledge, tools and processes associated with implementing and utilising a data warehouse. Candidates are given the skills to consolidate data from various sources into a centralised store, utilising ETL and ELT methods when applicable. Reporting tools are also demonstrated to teach candidates how to generate meaningful reports from the data warehouse.

Storing large amounts of data from various sources and generating valuable reports is an important skill for backend developers. Competencies from this course will enable candidates to contribute to data warehouse projects and utilise them to create value for future clients.

Course Learning Outcomes

Knowledge

The candidate:

- has knowledge of the concept and purpose of using data warehousing to analyse and report data
- has knowledge of theories and main approaches used to build a data warehouse system
- has knowledge of industry-relevant reporting tools used in data warehouse systems
- has insight into own opportunities for development in the field of data warehousing

Skills

The candidate:

- can find and refer to information relating to data warehousing
- can explain vocational choices when choosing an approach to build a data warehouse system
- can explain vocation choices when using tools to build reports
- can reflect over own data reporting and data warehouse system building and adjust it under supervision

General Competence
The candidate:

- can plan and carry out tasks aimed at building a data warehouse system alone or as part of a group and in accordance with ethical requirements and principles of data usage
- can plan and carry out tasks to generate reports with data extracted from a data warehouse
- can contribute to data warehouse system building and data reporting

Teaching and Learning - Campus
In this course, the following teaching and learning methods may be used on campus:

Lecture	Online Lessons
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online
In this course, the following teaching and learning methods may be used online:

Online Lessons	Lecture
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments
This is a list of requirements to pass the course:

Course Assignment	Pass/Fail

Course 4: Semester Project

Code	FI2BEP175	**NQF Level**	5.2
Credits	7.5	**Weeks**	4
Course Leader		N/A	
Hours Per Week		42 Hours	
Teacher-led hours		12	
Student-led hours		156	
Total Hours		168	

About the Course
The semester concludes with a graded project where the candidate must demonstrate practical skills and competence from courses in the first semester. Candidates work independently or in a group on a project, which must be planned, documented and executed according to project criteria. The aim is to carry out a practical project of elements from the previous courses to develop a web-based solution to a provided business case using an API.

The course builds on competence from *JavaScript in Practice, Database Technologies, Cloud Services*.

The project challenges candidates to use and combine accumulated competence from the previous courses and showcase how they can complete larger projects, either individually or across disciplinary boundaries. Candidates are challenged to think and work holistically, which provides a solid platform for further learning and understanding of the developer field.

Course Learning Outcomes
Knowledge
The candidate:

- can assess own work and the finished software solution in relation to the assignment description
- has insight into opportunities for development in planning and designing Back-end solutions

Skills
The candidate:

- can explain own vocational choices for tools, methods and techniques for planning, developing and design of Back-end solutions
- can reflect on own technical practices as a Back-end developer and adjust it under supervision
- can find and refer to information related to project work and assess its relevance to the project

General Competence
The candidate:

- can plan and carry out the development of Back-end solutions alone or as part of a group and in accordance with ethical requirements and principles
- can exchange points of view with others with a background in relation to the development of Back-end solutions

Teaching and Learning - Campus
In this course, the following teaching and learning methods may be used on campus:

Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online
In this course, the following teaching and learning methods may be used online:

Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments
This is a list of requirements to pass the course:

Semester Project	a grade from A-F

Course 5: Information Security

Code	FI2BEIS25	**NQF Level**	5.2
Credits	2.5	**Weeks**	2
Course Leader		N/A	
Hours Per Week		42 Hours	
Teacher-led hours		32	
Student-led hours		52	
Total Hours		84	

About the Course

This course provides candidates insight into the concepts, risks and regulations of information security. An overview of the various aspects of information security that concern Back-end development, together with an explanation of common risks and how to prevent them from being exploited, is provided. Privacy regulations, such as GDPR, are also covered in terms of their impact on secure and ethical Back-end solutions.

Information is a precious resource which needs to be protected effectively for strategic, ethical and regulatory reasons. There are considerable security risks involved in the building of software projects which need to be understood in order to prevent breaches. Candidates will have the relevant knowledge to practice due diligence against potential security threats.

Course Learning Outcomes

Knowledge

The candidate:

- has knowledge of the concepts and principles of information security
- has knowledge of the regulations, such as GDPR, governing data privacy
- has knowledge of the common security risks associated with Back-end applications
- is familiar with the importance of information security, GDPR and governing data privacy in the field of software and Back-end development

Skills

The candidate:

- can find and refer to information related to security-oriented project topics
- can reflect on own technical practices as a Back-end developer in terms of potential security risks and adjust it under supervision
- can find and refer to GDPR regulations and assess its relevance to a vocational software development project

General Competence

The candidate:

- can exchange points of view with others with a background in the trade/discipline and participate in discussions about compliance to information security concepts, best practices, and regulations
- can contribute to implementing the best practices for data governing in software development projects

Teaching and Learning - Campus

In this course, the following teaching and learning methods may be used on campus:

Lecture	Online Lessons
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online

In this course, the following teaching and learning methods may be used online:

Online Lessons	Lecture
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments

This is a list of requirements to pass the course:

Course Assignment	Pass/Fail

Course 6: JavaScript in Practice

Code	FI2BEJS75	**NQF Level**	5.2
Credits	7.5	**Weeks**	5
Course Leader		N/A	
Hours Per Week		42 Hours	
Teacher-led hours		80	
Student-led hours		130	
Total Hours		210	

About the Course

This course provides skills to utilise the advanced features of the JavaScript programming language to produce elegant algorithms for solving software problems. Candidates will be introduced to modern JavaScript features, functional programming concepts related to JavaScript, and common data structures.

Utilising the modern features of JavaScript is vital in producing maintainable and elegant software projects in practice. Candidates learn how to apply these features in a software solution effectively and collaborate with other developers in a structured manner using collaboration services.

Course Learning Outcomes

Knowledge

The candidate:

- has knowledge of the modern features of the JavaScript programming language
- has knowledge of commonly used algorithms and data structures used to solve software development-related problems
- has knowledge of best practices used at the workplace and in a business setting
- has knowledge of version control and collaboration services

Skills
The candidate:

- can explain the choice of utilising modern JavaScript features to develop solutions
- can explain the choice of algorithms and data structures used to solve software development-related problems
- can implement best practices used at the workplace
- can explain the choice of version control services

General Competence
The candidate:

- can plan and carry out software solutions in JavaScript in accordance with ethical requirements and principles for software development
- can exchange points of view with others with a background in relation to the development of elegant software solutions

Teaching and Learning - Campus

In this course, the following teaching and learning methods may be used on campus:

Lecture	Online Lessons
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online

In this course, the following teaching and learning methods may be used online:

Online Lessons	Lecture
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments

This is a list of requirements to pass the course:

Course Assignment	Pass/Fail

Course 7: Back-end Technologies

Code	FI2BEBT75	**NQF Level**	5.2
Credits	7.5	**Weeks**	6
Course Leader		N/A	
Hours Per Week		42 Hours	
Teacher-led hours		96	
Student-led hours		156	
Total Hours		252	

About the Course

This course provides the skills needed to create a robust Back-end solution using the .NET Framework coupled with the C# programming language. Candidates will be given the knowledge needed to utilise basic data types, data structures and control structures required to produce a Back-end solution. Widely used tools and Back-end specific frameworks will also be covered, such as ASP.NET and Entity Framework, which streamline development.

Familiarity with frameworks, libraries and tools is essential to developing most modern Back-end software projects when productivity is considered. This course provides candidates with the essential knowledge needed to build a Back-end solution with the .NET framework and its associated tools. It also enables the candidate to transition to a similar framework with the knowledge of the high-level concepts it is based on.

Course Learning Outcomes

Knowledge
The candidate:

- has knowledge of the programming language C# used for developing Back-end applications using the .NET Framework
- has knowledge of the industry-relevant tools such as ASP.NET and Entity Framework used in the development of Back-end .NET applications

Skills
The candidate:

- can explain the functionality and architecture behind the .NET framework as it relates to Back-end software projects
- can find and refer to technical information about the .NET framework and assess its relevance for each specific project

General Competence
The candidate:

- can plan and carry Back-end software projects using C# and the .NET Framework alone or as part of a group
- can exchange points of view with others with a background in relation to the development of .NET-based projects

Teaching and Learning - Campus

In this course, the following teaching and learning methods may be used on campus:

Lecture	Online Lessons
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online

In this course, the following teaching and learning methods may be used online:

Online Lessons	Lecture
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments

This is a list of requirements to pass the course:

Course Assignment	Pass/Fail

Course 8: Front-end Technologies

Code	FI2BEFT75	**NQF Level**	5.2
Credits	7.5	**Weeks**	5
Course Leader		N/A	
Hours Per Week		42 Hours	
Teacher-led hours		80	
Student-led hours		130	
Total Hours		210	

About the Course

This course instructs candidates on utilising Front-end libraries, frameworks and tools to create robust, maintainable user interfaces. Candidates will also be taught how to utilise various tools and complementary technologies commonly used in conjunction with the JavaScript framework.

Front-end JavaScript frameworks and libraries are widely used in modern Full-stack software projects in order to streamline the development of modular user interfaces. The candidate will learn how to use such frameworks and libraries to contribute to Front-end tasks themselves as a Full-stack developer and collaborate easily with Front-end developers as Back-end developers.

Course Learning Outcomes

Knowledge

The candidate:

- has knowledge of widely used JavaScript Front-end frameworks, libraries and tools to build Full-stack applications
- has knowledge of component-based architecture used to create modular Front-end applications
- has knowledge of web sockets to deal with real-time communication in a Full-stack application
- is familiar with the history and traditions of Full-stack application development
- has insight into own opportunities to become a Full-stack software developer using JavaScript frameworks, libraries and tools

Skills

The candidate:

- can explain vocational choices in implementing a Full-stack solution using JavaScript frameworks, libraries and tools
- can explain vocational choices with modular components
- can explain vocational choices to implement real-time communication in a Full-stack application
- can reflect on own vocational practice developing a Full-stack application and adjust it under supervision

General Competence

The candidate:

- can plan and carry out tasks building a full stack application using JavaScript frameworks, libraries and tools alone or as part of a group
- can exchange points of view with others with a background in the trade/discipline and participate in discussions about the development of good practice
- can exchange points of view with Front-end developers and participate in discussions about the development of good Full-stack developer practice

Teaching and Learning - Campus

In this course, the following teaching and learning methods may be used on campus:

Lecture	Online Lessons
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online

In this course, the following teaching and learning methods may be used online:

Online Lessons	Lecture
Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments

This is a list of requirements to pass the course:

Course Assignment	Pass/Fail

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Course 9: Exam Project

Code	FI2BEP275	**NQF Level**	5.2
Credits	7.5	**Weeks**	5
Course Leader		N/A	
Hours Per Week		42 Hours	
Teacher-led hours		15	
Student-led hours		195	
Total Hours		210	

About the Course

This major project reflects the competence the candidates have acquired during the academic year.

The candidate must solve the assignment independently or as part of a group from a practical problem. The candidate is responsible for all aspects of the project. The academic staff will present alternative case projects. The completed project must be uploaded through appropriate channels.

The course builds on competence from all courses during the academic year.

The course aims to provide candidates with the ability to make independent choices and deliver a comprehensive product from start to finish with professional standards, deadline compliance and focus on efficient workflows. The candidate must demonstrate the ability to make reasonable and efficient choices through the use of software tools and detailed documentation, in addition to communicating with professional terminology and expressions during a project.

Course Learning Outcomes

Knowledge

The candidate:

- can assess own work and the accomplished software solution in relation to the assignment description or project brief
- has insight into opportunities for development in planning and designing Back-end solutions

Skills

The candidate:

- can explain own vocational choices for tools, methods and techniques for planning, developing and design of Back-end solutions
- can reflect on own technical practices as a Back-end developer and adjust it under supervision
- can find and refer to information related to project work and assess its relevance to the project

General Competence

The candidate:

- can plan and carry out the development of Back-end solutions individually or as part of a group and in accordance with ethical requirements and principles for software development
- can exchange points of view with others with a background in relation to the development of Back-end solutions

Teaching and Learning - Campus

In this course, the following teaching and learning methods may be used on campus:

Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Teaching and Learning – Online

In this course, the following teaching and learning methods may be used online:

Guidance	Self-study

Where applicable, teaching materials, reading lists, and essential resources will be shared in the learning platform and software user manuals.

Work requirements and Assessments

This is a list of requirements to pass the course:

Exam Project	a grade from A-F